RAGHUL SAYEE KUPPA ANANDHACHARY DHINAKARAN

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SUMMARY

Software Engineer and MSCS candidate at Northeastern University with 2+ years of experience building GenAI applications, agentic systems, and scalable backend services. Co-inventor of an enterprise agentic-AI platform.

EXPERIENCE

Senior Associate Software Engineer, Light & Wonder — **Aug 2023 - Jul 2025**

- Designed and developed Knowledge Retrieval, Code Assistance, and Log Analysis agents using locally deployed LLMs, RAG, and tool-enabled workflows for grounded enterprise search, AI-assisted development, and automated system-log diagnostics, improving team productivity by 40%.
- Fine-tuned and optimized open-source LLMs using Unsloth and deployed them through Ollama for secure, low-latency local inference.
- Built and optimized full-stack product features using Angular, .NET 8.0, and SQL Server, improving response times by 60%+.
- Designed and deployed secure REST APIs with role-based access control, strengthening authorization and reducing unauthorized-access risk.
- Engineered a GPU-accelerated facial recognition system using DeepFace, OpenCV, Flask APIs, Angular, and Raspberry Pi for real-world edge deployment.
- Automated CI/CD pipelines with Jenkins, cutting release cycles by 40% and improving deployment reliability.

Full Stack Java Developer Intern, Light & Wonder — **Feb 2023 - Jul 2023**

- Built a Casino Game Guide web application using Angular + Spring Boot, enhancing player engagement and navigation.
- Optimized APIs and UI workflows, improving system performance and user experience.

Data Science Intern, MIT Square Services Pvt. Ltd. — **Dec 2021 - Jan 2022**

- Developed ML models for student performance prediction using Python + scikit-learn, achieving 92% accuracy.
- Applied data cleaning, feature engineering, and evaluation pipelines to deliver reliable predictive insights.

PATENTS

Agentic Artificial Intelligence Gaming Platform - Co-Inventor, U.S. Patent Application, Attorney Docket P7969-US-ORG, 2025

SKILLS

- **AI Product Engineering:** Generative AI, RAG, LLM Fine-tuning, Prompt Engineering, LLM-based Agent Workflows, Personalization, AI-powered Workflows, LangChain, LangGraph, Vector Databases
- **Frontend:** React, Angular, TypeScript, JavaScript, HTML, CSS, Responsive UI, Client-Server Architecture
- **Backend & APIs:** Python, Flask, Spring Boot, C#, .NET 8.0, REST APIs, API Integration, Microservices, Backend Services
- **Scalable Systems & Cloud:** AWS, Docker, Jenkins, CI/CD, Distributed Systems, Scalable Applications, Linux, Git, Postman
- **ML/DL:** Machine Learning, Deep Learning, NLP, Computer Vision, TensorFlow, PyTorch, scikit-learn, OpenCV, YOLO, OCR
- **Programming Languages:** Python, JavaScript, TypeScript, Java, C#, C++, R
- **Databases & Data:** SQL, SQL Server, Apache Spark, Hadoop, MapReduce, Tableau
- **CS Fundamentals:** Data Structures & Algorithms, Object-Oriented Programming, Problem Solving

PUBLICATION

Traffic Violation Detection
8th National Conference on Advanced Computing Technologies (NCACT'23), IJIRSET, April 2023.

ACHIEVEMENTS

Hackathon Winner – Special Category Award
L&W SYSrupt Season 4 Hackathon for teamwork and innovative solutions.

PROJECTS

Multimodal AI Lifestyle Assistant with Agentic Planning and Personalization

- Built a full-stack AI assistant using RAG, LLM agents, vector search, multimodal inputs, and API integrations to deliver personalized planning, recommendations, product comparisons, and everyday task support.
- Designed backend workflows for intent detection, task decomposition, memory retrieval, prompt orchestration, tool calling, response validation, and evaluation, with React/TypeScript UI components for conversational experiences, dynamic cards, comparison views, and feedback-driven refinement.

Real vs. Fake Audio Detection

- Developed an audio deepfake detection system using Logistic Regression, SVM, XGBoost, and CNN models on ASVspoof 2021 DF and Deepfake-Eval-2024 datasets.
- Designed a hybrid dual-branch model combining MFCC and spectral features with CNN-learned log-mel spectrogram embeddings, evaluated using F1-score, ROC-AUC, and confusion matrices.

Traffic Violation Detection

- Designed a YOLOv4 + CNN + OCR pipeline for automated traffic violation detection and vehicle identification; demonstrated real-time classification and reporting.
- Published research at NCACT'23 (IJIRSET).

EDUCATION

Master of Science in Computer Science — **May 2027 (expected)**
Northeastern University, Boston — **GPA : 3.5/4.0**

Bachelor of Engineering in Computer Science And Engineering — **Apr 2023**
Anna University, Chennai. — **GPA : 9.11/10**